Abdelilah AITROUGA

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Abdelilah Aitrouga | Abdo-RG

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Profile

PhD student in **Artificial Intelligence** at **Mohammed VI Polytechnic University (UM6P)** working on multimodal learning and neural decoding. My research focuses on bridging **brain activity and visual representations** using deep generative models.
I work on large-scale fMRI datasets and scalable deep learning architectures to reconstruct and interpret visual information from neural signals. Previously, I worked on **text-driven video editing**, **video generation**, and **video understanding**.
I hold a Master's degree in **Data Science & Big Data** from **ENSIAS (ranked #1)** and a Bachelor's degree in **Applied Mathematics**.

Education

- **Mohammed VI Polytechnic University** — *Septembre 2023 - Septembre 2027*
  PhD in Artificial Intelligence, AI Movement – International Artificial Intelligence Center of Morocco — Rabat, Morocco
- **Mohammed V University** — *Septembre 2021 - July 2023*
  Master's Degree, ENSIAS (Majorer #1) — Rabat, Morocco
- **Ibn Zoher University** — *Septembre 2018 - July 2021*
  Bachelor's Degree, FSA — Agadir, Morocco
- **Abdellah Ibn Yassine College** — *Septembre 2017 - July 2018*
  Baccalaureate Degree, Mathematics Science -A) — Inzegane, Morocco

Publications

[C1] A. Aitrouga, et al. (2025). **Swin-Editor: Enhancing Consistency in Text-Driven Video Editing**. *International Conference on Artificial Intelligence and Soft Computing (ICAISC)*, Springer. doi:10.1007/978-3-032-03708-4_2

[Preprint] A. Aitrouga, et al. (2025). **VRWKV-Editor: Reducing Quadratic Complexity in Transformer-Based Video Editing**. *arXiv preprint*. https://doi.org/10.48550/arXiv.2509.25998

Projects

- **Video Classification — Deep Learning–Based Action Recognition** — *2023*
  Tools: Python, PyTorch, OpenCV, NumPy, Scikit-learn []
  - Developed a video classification pipeline using CNN–LSTM architectures to recognize human actions from short video clips
  - Implemented efficient video frame extraction and preprocessing using OpenCV to improve temporal feature quality
  - Created a modular training framework with support for custom datasets, boosting reproducibility and scalability
  - Applied statistical evaluation methods to analyze class performance and model generalization
- **CBIR (Content-Based Image Retrieval) — Image Similarity Search System** — *2022*
  Tools: Python, OpenCV, Scikit-learn, Matplotlib []
  - Developed a CBIR system extracting color histograms, texture descriptors, and shape features to compare and retrieve similar images
  - Implemented feature vector indexing and distance-based retrieval, supporting large-scale image collections
  - Created interactive visualizations to inspect ranked similarity results and feature-space distributions
  - Developed reusable modules for feature extraction and similarity scoring, enabling easy integration into other systems

Technical Skills

- **Programming** Python, PyTorch, NumPy
- **Machine Learning** Deep Learning, Transformers, Diffusion Models, Multimodal Learning
- **Computer Vision** Video Understanding, Image Retrieval, Visual Representation Learning
- **Tools** OpenCV, Scikit-learn, Git, Linux, SLURM
- **Data** Large-scale datasets, fMRI data processing